

Patrick O'Brien

Identity & Access Management | Cloud Automation | Microsoft Entra ID / Azure

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PROFESSIONAL SUMMARY

Entra ID Global Administrator focused on identity and access automation, with a B.S. in Computer Science (2025), managing Microsoft 365, Exchange, MFA, and Conditional Access in a 400+-user, 100+-property tenant. Builds production Python tooling on the Microsoft Graph API that automates the Graph side of onboarding and offboarding, with scoped app-only permissions for directory writes, dry-run mode, and an audit log of completed actions. Software engineering background in REST API design, data modeling, and CI/CD applied to identity problems.

SKILLS

Identity & Cloud: Microsoft Entra ID (Azure AD), multi-factor authentication (MFA), Conditional Access, app registrations and service principals, Microsoft Graph API, Microsoft 365, Exchange Online, Azure, Azure Key Vault, user lifecycle management (JML), least-privilege access, role-based access control, OIDC workload identity federation, audit logging

Automation & Scripting: Python, PowerShell, REST API integration, CLI tooling, YAML configuration

DevOps & Tooling: Terraform, infrastructure as code, Git/GitHub, GitHub Actions (CI/CD), Docker, pytest, automated testing

Software Engineering: TypeScript, JavaScript, SQL, Java, C++, FastAPI, React, SQLAlchemy, PostgreSQL, MySQL, JWT auth

EXPERIENCE

IT Support Specialist | Jun 2025 – Present

Habitat America, Annapolis, MD

- Independently designed and built Python tools on the Microsoft Graph API, adopted in production with manager approval, that automate Entra ID provisioning and deprovisioning in a 400+-user, 100+-property tenant, replacing manual per-account setup with a single command.
- Serve as Entra ID Global Administrator for 400+ users across 100+ properties, managing MFA, Conditional Access, account lifecycle, license assignment, group membership, and access permissions across Microsoft 365 and Exchange.
- Created and manage the Entra app registrations behind these tools: app-only (client-credentials) Graph API authentication with scoped application permissions for all directory writes, dry-run mode, and an audit log of completed actions.
- Diagnose and resolve system and application issues under time constraints across a distributed property portfolio.

Operations & Database Manager | Mar 2019 – May 2025

Mike's Mowing (family business), Ridgely, MD

- Designed and maintained a Microsoft Access database for customer records, service history, and scheduling over 6 years.

Job Coach | Mar 2013 – Mar 2019

Benedictine, Ridgely, MD

- Provided on-the-job coaching and support for individuals with disabilities in workplace settings.

PROJECTS

Entra ID Infrastructure as Code · Terraform, Azure, Entra ID, Key Vault, GitHub Actions (OIDC)

github.com/pobriendev/entra-terraform

- Codified the identity layer my Graph API tools need as Terraform in a personal sandbox tenant: app registration, scoped Graph application permissions, service principal role assignments, and a 180-day client secret stored in Key Vault, replaced on the next apply.
- Solved the remote-state bootstrap problem with a separate root module that creates the state account with Entra-only auth, shared-key access disabled, versioning and soft delete.
- Built a GitHub Actions plan-on-PR pipeline authenticating via OIDC workload identity federation with no stored secret, using a CI identity with no Azure write roles and redacted plan output for a public repository.

Employee Provisioning Tool · Python, Microsoft Graph API, Entra ID, YAML

github.com/pobriendev/employee-provisioning-tool

- Python CLI automating Entra ID onboarding and offboarding: account creation or role-account reuse, license assignment, group membership, and credential delivery, replacing a manual portal process.
- App-only Graph authentication for directory writes, dry-run mode, lockout-first offboarding, and an audit log of completed actions.

Entra Stale Accounts · Python, Microsoft Graph API, PyPI, GitHub Actions

github.com/pobriendev/entra-stale-accounts

- Published a read-only CLI to PyPI (pip install entra-stale-accounts) flagging inactive Entra ID accounts past a configurable threshold, including never-signed-in accounts — an access-hygiene and audit tool.
- Generalized from production automation into a standalone package with a mocked test suite, GitHub Actions CI, and Graph Retry-After handling.

Ticket Management System · FastAPI, PostgreSQL, React, JWT, Docker, GitHub Actions

github.com/pobriendev/ticket-system

- Helpdesk ITSM application with role-based access control, query-level visibility scoping, an enforced ticket status lifecycle, priority-derived SLA due dates, and an automatic audit log of every ticket field change.
- 175+ backend tests (~96% coverage; CI fails under 85%) and 135+ frontend tests, CI-gated. Live: ticket-system-azure-omega.vercel.app.

EDUCATION

Delaware State University — B.S. in Computer Science (GPA 3.57), 2025

Delaware Technical Community College — A.S. in Computing and Information Science, 2020

CERTIFICATIONS

Microsoft Certified: Identity and Access Administrator Associate (SC-300) — in progress